

STUDENT ACTIVITY FILE

INNOVATIVE DESIGN LAB - I

Academic year: 2026-2027

Department: Computer Science and Engineering

Section: Interdisciplinary

Innovation Ecosystem Project ID: SAI-IDL1-2026-CSE-09

Title of the Project: Autonomous Hybrid-AI SOC Analyst & Investigative Engine

Inter disciplinary Project : ☒ Yes ☐ No **Team**

Members (Max. 3)

S.No	Name	Student ID
1	Gokula Kannan M (Team Lead)	SEC25CS196 (CSE-A)
2	Lakshan M	SEC25CS036 (CSE-A)
3	Sivabalan T	SEC25CS101 (CSE-C)

Project Version Is this project:

☐ Version 1 – New Project ☐ Version 2–Pass out Student Project(Carried Forward)

(If Version 2, provide the following details of the original project):

Previous Project Title	Sairam Innovation Project ID

Domain Name: Cybersecurity, Artificial Intelligence & SIEM Automation

IEEE Society: IEEE Computer Society / IEEE Cybersecurity Community

IEEE Community: IEEE Student Branch Sairam

Club and Cells: Sairam Innovation Ecosystem / Cyber Security Club

Name of the department IDL - I Coordinator: Dr. A. SHEELA (Associate Professor) **Faculty ID:** IDL-FAC-01

Name of the Supervisor: Dr. A. SHEELA (Associate Professor) **Faculty ID:** IDL-FAC-02

Name of the External Guide: Senior Tech Leads & Cyber Forensic Specialists

Designation & Organization: Kerala Police Cyberdome / Zoho Corporation

SDG	Goal Number With Name	SAP Code with Explanation
Primary SDG:	SDG 9: Industry, Innovation, and Infrastructure	1. 2. 3.
Secondary SDG:	SDG 16: Peace, Justice, and Strong Institutions	1. 2. 3.
Tertiary SDG:	SDG 8: Decent Work and Economic Growth	1. 2. 3.

S.No.	Title	Start Date	Completion Date	Mark (Out of 10)	Remarks	Signature of the IDL - I Coordinator
1	Orientation & Entrepreneurial Mindset	01/08/2026	07/08/2026	10	Completed	
2	Problem Identification & Passion CV	08/08/2026	14/08/2026	10	Completed	
3	Customer Segmentation & Persona Creation & JTBD	15/08/2026	21/08/2026		In Progress	
4	Ideation, Market Size & Competitor Analysis	22/08/2026	28/08/2026		Upcoming	
5	Milestone 1: Problem-Solution Fit Presentation	29/08/2026	04/09/2026		Upcoming	
6	MVP Design & Prototyping					
7	Opportunity Assessment & Feasibility					
8	Business Model & Lean Canvas					
9	Literature survey					



10	Milestone 2 : Business Model Presentation					
11	Marketing & Sales Strategy (GTM)					
12	Financial Planning & Team Building					
13	Milestone 3 : Final Pitch- Venture Feasibility					

Outside World Projection

Details of the Competition Attended (Hackathon / Idea Pitching)

S.No	Name	Date	Place	Remark	Signature
1	Hac'KP 2026 (7th Edition National Hackathon by Kerala Police Cyberdome)	14/08/2026	Zoho Corporation, Chennai		Selected for National In-Person Pitching Round
2					

Design Patent / Idea Patent Publication / Conference Paper

1	Confidentiality-Preserving Autonomous SOC Triage via Tokenized Edge-Cloud Multi-Agent Systems	August 2026	IEEE / USENIX Track		Paper Manuscript & Patent Drafted
2					

IDL – I Coordinator
Dr. A. SHEELA

Creative Innovator
Gokula Kannan M (Team Lead)

H.O.D
Head of Department



புயன்றால் புதுயாதது
எதுவும் இல்லை.

சென்னை
M. Ramesh

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PHOTOS / LITERATURE PAPERS / PATENT SEARCH / SKETCHES (PASTE OR ATTACH SEPARATE PAGES)

WEEK – 1

1. Orientation & Entrepreneurial Mindset

Security Operations Centers (SOCs) generate over 10,000 security logs daily, creating severe alert fatigue where human analysts spend 30-45 minutes manually triaging single events. Commercial AI endpoints cannot be deployed directly due to data privacy regulations and risks of uploading confidential network payloads to external servers.

- **Understand entrepreneurship & innovation**

The proposed solution, SENTINEL, introduces Reversible Tokenized Pseudonymization, scrubbing 100% of sensitive PII, IP addresses, and hostnames locally before AI reasoning occurs. This enables enterprise-grade autonomous triage at \$0 operational cost for 90% of routine alerts using local consumer GPU hardware (NVIDIA RTX 3050).

The global SIEM automation and threat intelligence market is valued at \$13.2 Billion+ with an 18.5% CAGR. Primary target markets include Enterprise SOCs, Managed Security Service Providers (MSSPs), Air-Gapped Defense Networks, Cyber Crime Units, and Startups.

- **opportunity recognition, entrepreneurial traits and growth mindset** Technological security needs can be met by quantizing LLM parameters and is now with 4-bit (Q4_K_M) GGUF format to execute high-capacity intelligence on edge workstations.

- Startup journey from idea to venture.

1. Problem Formulation: Identifying SOC alert fatigue and data privacy constraints.
2. Proof-of-Concept: Engineering local Zero-Trust Data Sanitizer (src/sanitizer.py).
3. MVP Architecture: Building 3-Tier Hybrid AI Router (src/router.py).
4. Validation & Pitching: Selected for National Pitching Round at Hac'KP 2026 @ Zoho Corporation.
5. Commercialization: Deploying open-source SaaS framework and enterprise MSSP engine.

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WEEK – 2

Modern digital investigations face a severe bottleneck where human investigation speed cannot match machine-speed cyber attacks. Public commercial cloud LLMs violate legal data privacy and chain of custody mandates when processing un-sanitized evidence.

2. Problem Identification & Passion CV

- Identify real-world problems based on personal interests

Small businesses, hospitals, and educational institutions are unable to afford \$100k+/year enterprise SIEM subscriptions (e.g., Splunk or Cortex XSOAR), leaving critical infrastructure vulnerable to ransomware. SENTINEL addresses this societal gap by delivering a self-hosted,

- Skills, experiences, and societal needs

- Prepare a Passion CV to identify potential entrepreneurial opportunities.

PASSION CV & TEAM PROFILE MATRIX:

? Gokula Kannan M (Team Lead ? SEC25CS196, CSE-A Core):

Domain: Software Architecture, Systems Engineering, Project Governance & Venture Strategy.

Contribution: Overall system architecture, sprint management, and leading national pitch at Hac'KP 2026 @ Zoho.

? Lakshan M (Team Member 1 ? SEC25CS036, CSE-A Core):

Domain: Full-Stack Web Development, React.js UI, WebSockets Integrations.

Contribution: Real-time incident dashboard engineering and Human-in-the-Loop (HITL) action approval interface.

? Sivabalan T (Team Member 2 ? SEC25CS101, CSE-C Core):

Domain: AI Engine Architecture, Zero-Trust Privacy Engineering, Quantization & Vector RAG.

Contribution: Development of Zero-Trust Data Sanitizer (src/sanitizer.py) and 3-Tier AI Router (src/router.py).

? Faculty Supervisor: Dr. A. SHEELA (Associate Professor, Dept. of Computer Science & Engineering).